

Tutorial 6: Column Oriented Databases

Question 1

Prepare a column-oriented representation for the following row-oriented table.

TranID	TranDate	ProductCode	Category	CustNo	Qty	TotalAmt	Discount
7582	2023-08-01	GT4235	Media	V899	7	350.00	10%
7431	2023-08-01	HT8865	Food	N776	24	150.00	0%
9921	2023-08-01	RT5412	Gadget	V899	5	250.00	10%
4143	2023-08-02	GT1245	Media	N998	100	1200.00	0%
8876	2023-08-02	TR554	Food	V665	12	24.00	10%
9900	2023-08-02	GT4235	Media	V665	50	2500.00	10%
6548	2023-08-02	HT8865	Food	V665	24	150.00	10%

Give an example of an SQL query that will perform ‘significantly’ better with the column-oriented representation.

Question 2

Match the following compressions for the most suitable column in the provided example in Q1:

- a) Dictionary Encoding
- b) Bit-Packing Encoding -T64
- c) Delta Encoding

Conduct the respective compressions and explain why the match you propose is appropriate.

Question 3

Explain how query performance is optimized using:

- a) Sparse Primary Index
- b) Materialised Views
- c) Projections

---ooo0000000ooo---